

Week 3: The Avellaneda-Stoikov Model

Optimal market making via stochastic control – derived from scratch

North Carolina State University | Summer 2026

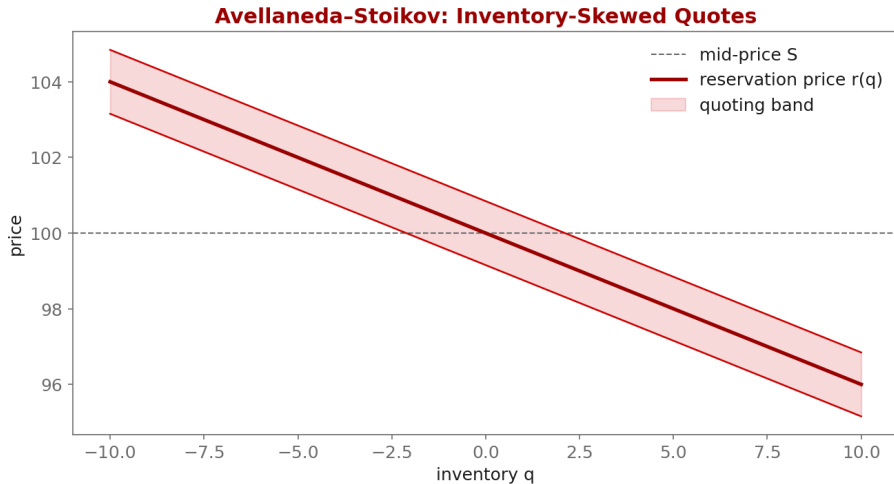
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- ▶ Set up the Avellaneda-Stoikov control problem
- ▶ Follow the HJB derivation and understand every modeling choice
- ▶ Interpret the reservation price and the optimal half-spread

- ▶ Mid-price: $dS_t = \sigma dW_t$ (arithmetic Brownian motion)
- ▶ Fill intensity: $\lambda(\delta) = A \exp(-\kappa \delta)$
- ▶ Wealth from fills; inventory q jumps by ± 1 on a fill
- ▶ Exponential (CARA) utility of terminal wealth

- ▶ Dynamic programming $\rightarrow$ HJB partial differential equation
- ▶ Separation ansatz collapses HJB to ODEs in inventory
- ▶ Asymptotic (long-horizon) closed-form quotes
- ▶ Reservation price skews quotes against inventory



$$dS_t = \sigma dW_t$$

Why: Arithmetic (not geometric) Brownian motion is chosen because over a short market-making horizon price changes are small and roughly additive; it keeps the HJB linear in S and makes the cash dynamics exact.

$$\lambda(\delta) = A e^{-\kappa \delta}$$

Why: Fill intensity decays with quote distance δ . The exponential form is the simplest law consistent with 'further from mid $\Rightarrow$ fewer fills'; A sets the base rate, κ the decay. It also yields a closed-form optimum.

$$\max \mathbb{E} \left[-e^{-\gamma (X_T + q_T S_T)} \right]$$

Why: CARA (exponential) utility is used because constant absolute risk aversion makes the value function factor out wealth, so the optimal quotes do not depend on the cash level – exactly what we want for a market maker.

$$\partial_t V + \frac{1}{2} \sigma^2 \partial_{SS} V + \max_{\delta^b} \lambda(\delta^b) [V(q+1, x-S+\delta^b) - V] + \max_{\delta^a} \lambda(\delta^a) [V(q-1, x+S+\delta^a) - V] = 0$$

Why: The HJB equation: the diffusion term prices inventory risk; each max term is the dealer optimally choosing the bid/ask offset, trading fill probability against captured spread.

$$V(t, x, S, q) = -e^{-\gamma(x+qS)} \theta(t, q)$$

Why: This ansatz is not a guess from nowhere: under CARA + arithmetic dynamics the value function inherits a multiplicative structure across (x, qS) , so all the wealth/price dependence is exponential and only $\theta(t, q)$ remains to solve.

$$r(t, q) = S - q \gamma \sigma^2 (T - t)$$

Why: The reservation (indifference) price: it sits BELOW mid when you are long ($q > 0$) so you quote to sell down inventory. The skew scales with risk aversion γ , variance σ^2 , and time left – more risk $\Rightarrow$ more skew.

$$\delta^* = \frac{1}{\gamma} \ln\left(1 + \frac{\gamma}{\kappa}\right) + \frac{\gamma\sigma^2}{2}(T - t)$$

Why: The optimal half-spread = an order-flow term $\frac{1}{\gamma} \ln(1 + \gamma/\kappa)$ (how hard fills are to get) plus an inventory-risk term $\frac{\gamma\sigma^2}{2}(T - t)$. You widen when risk-averse, when volatile, and when far from the horizon.

- ▶ This week uses simulated dynamics (no live data needed).
- ▶ For realistic sigma, estimate it from Coincall perp klines:
 - ▶ GET /open/option/market/kline/history/v1/{name}?period=m1 (or futures klines)
- ▶ You will calibrate A, kappa from real fills in Week 5.

Cross-check exact paths at <https://docs.coincall.com/>

```
import torch

def as_quotes(S, q, gamma, sigma, kappa, tau):
    """Asymptotic Avellaneda-Stoikov bid/ask. Vectorized over inventory q."""
    reservation = S - q * gamma * sigma**2 * tau
    half_spread = (1.0/gamma) * torch.log1p(gamma/kappa) \
        + 0.5 * gamma * sigma**2 * tau
    return reservation - half_spread, reservation + half_spread # bid, ask

S = torch.tensor(100.0); q = torch.arange(-10, 11).float()
bid, ask = as_quotes(S, q, gamma=0.1, sigma=2.0, kappa=1.5, tau=1.0)
print("inventory ->", q.tolist())
print("bid ->", bid.round(decimals=2).tolist())
```

- ▶ Avellaneda-Stoikov (2008); Cartea-Jaikumar-Penalva, Ch. 10